**North South University**
Department of Electrical & Computer Engineering

# ERD AND SCHEMA DESIGN

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|---|---|
| **Course Code** | CSE311L |
| **Course Title** | Database Systems Lab |
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| **Section** | 05 |
| **Group Number** | 6 |
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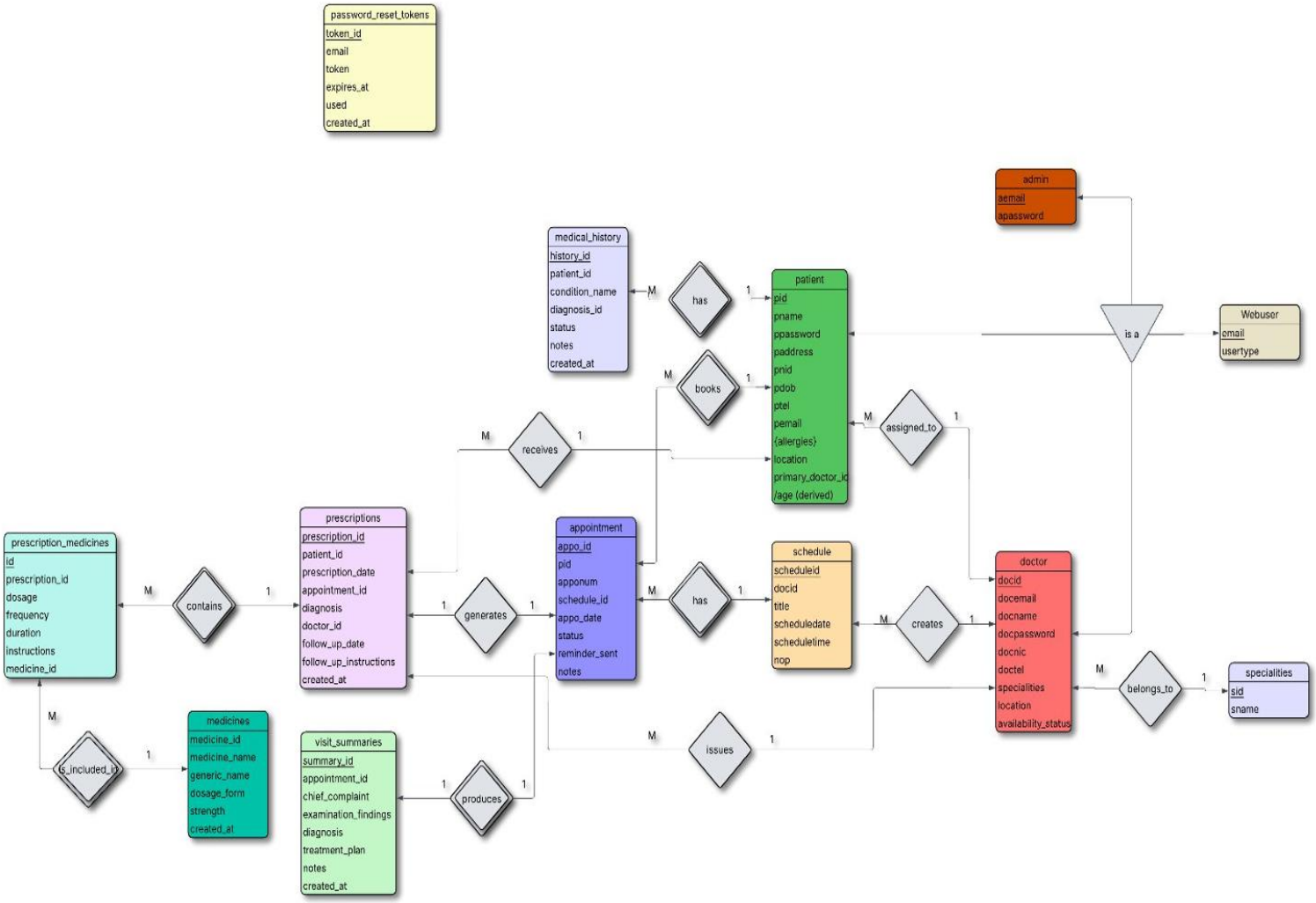
# ER Diagram and Schema Design

## ERD



# Entities & Attributes (with Primary Keys)

## Patient

- **pid (PK)**
- pname
- ppassword
- paddress
- pnid
- pdob
- ptel
- pemail
- allergies
- location
- primary_doctor_id (FK)
- age (derived)

## Doctor

- **docid (PK)**
- docemail
- docname
- docpassword
- docnic
- doctel
- specialities (FK)
- location
- availability_status

## Appointment

- **appo_id (PK)**
- pid (FK)
- apponum
- schedule_id (FK)
- appo_date
- status
- reminder_sent
- notes

## Schedule

- **scheduleid (PK)**

- docid (FK)
- title
- scheduledate
- scheduletime
- nop

## Prescriptions

- **prescription_id (PK)**
- patient_id (FK)
- prescription_date
- appointment_id (FK)
- diagnosis
- doctor_id (FK)
- follow_up_date
- follow_up_instructions
- created_at

## Prescription_Medicines

- **id (PK)**
- prescription_id (FK)
- dosage
- frequency
- duration
- instructions
- medicine_id (FK)

## Medicines

- **medicine_id (PK)**
- medicine_name
- generic_name
- dosage_form
- strength
- created_at

## Visit_Summaries

- **summary_id (PK)**

- appointment_id (FK)
- chief_complaint
- examination_findings
- diagnosis
- treatment_plan
- notes
- created_at

## Medical_History

- **history_id (PK)**
- patient_id (FK)
- condition_name
- diagnosis_id
- status
- notes
- created_at

## Specialities

- **sid (PK)**
- sname

## Admin

- **aemail (PK)**
- apassword

## WebUser

- **email (PK)**
- usertype

## Password_Reset_Tokens

- **token_id (PK)**
- email (FK)
- token
- expires_at
- used
- created_at

## Relationships & Cardinality

Patient - Medical_History: 1:M
Patient - Appointment: 1:M
Patient - Doctor: M:1
Doctor - Schedule: 1:M
Schedule - Appointment: 1:M
Appointment - Prescription: 1:1
Prescription - Patient: M:1
Prescription - Prescription_Medicines: 1:M
Medicines - Prescription_Medicines: 1:M
Appointment - Visit_Summary: 1:1
Appointment - Doctor: M:1
Doctor - Specialities: M:1
Admin & Patient -> WebUser: IS-A

## Foreign Keys

Patient.primary_doctor_id -> Doctor.docid
Appointment.pid -> Patient.pid
Appointment.schedule_id -> Schedule.scheduleid
Schedule.docid -> Doctor.docid
Prescriptions.patient_id -> Patient.pid
Prescriptions.appointment_id -> Appointment.appo_id
Prescriptions.doctor_id -> Doctor.docid
Prescription_Medicines.prescription_id -> Prescriptions.prescription_id
Prescription_Medicines.medicine_id -> Medicines.medicine_id
Visit_Summaries.appointment_id -> Appointment.appo_id
Medical_History.patient_id -> Patient.pid
Doctor.specialities -> Specialities.sid
Password_Reset_Tokens.email -> WebUser.email

# Logical Schema

Patient(pid [PK], pname, ppassword, paddress, pnid, pdob, ptel, pemail, allergies, location, primary_doctor_id [FK → Doctor], age)

Doctor(docid [PK], docemail, docname, docpassword, docnic, doctel, specialities [FK → Specialities], location, availability_status)

Appointment(appo_id [PK], pid [FK → Patient], apponum, schedule_id [FK → Schedule], appo_date, status, reminder_sent, notes)

Schedule(scheduleid [PK], docid [FK → Doctor], title, scheduledate, scheduletime, nop)

Prescriptions(prescription_id [PK], patient_id [FK → Patient], prescription_date, appointment_id [FK → Appointment], diagnosis, doctor_id [FK → Doctor], follow_up_date, follow_up_instructions, created_at)

Prescription_Medicines(id [PK], prescription_id [FK → Prescriptions], dosage, frequency, duration, instructions, medicine_id [FK → Medicines])

Medicines(medicine_id [PK], medicine_name, generic_name, dosage_form, strength, created_at)

Visit_Summaries(summary_id [PK], appointment_id [FK → Appointment], chief_complaint, examination_findings, diagnosis, treatment_plan, notes, created_at)

Medical_History(history_id [PK], patient_id [FK → Patient], condition_name, diagnosis_id, status, notes, created_at)

Specialities(sid [PK], sname)

Admin(aemail [PK], apassword)

WebUser(email [PK], usertype)

Password_Reset_Tokens(token_id [PK], email [FK → WebUser], token, expires_at, used, created_at)